

Math 118– Lecture 02

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1 Norms, Inner Products, Cauchy, etc

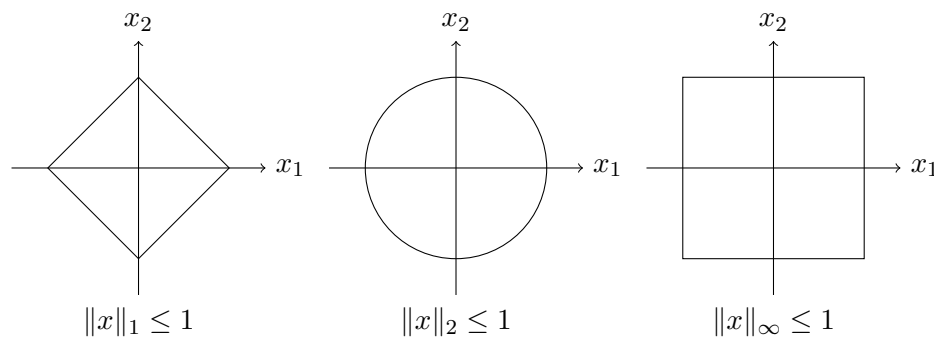
Definition 1.1 (norm). A norm on a vector space V is a mapping $\|\cdot\| : V \rightarrow [0, \infty)$. with properties

- Triangle Inequality: $\|u + v\| \leq \|u\| + \|v\|$.
- Homogeneity $\|\alpha u\| = |\alpha| \|u\|$
- Positive Definiteness. $\|u\| = 0 \iff u = 0$
- Distance from u and v is $\|u - v\|$

1.1 Norms on R^n or C^n

- One-norm: $\|x\|_1 = \sum_{k=1}^n |x_k|$
- Two-norm: $\|x\|_2 = \sqrt{\sum_{k=1}^n (x_k)^2}$
- Max-norm: $\|x\|_\infty = \max_{k \leq n} |x_k|$

Definition 1.2 (Unit Ball). $B = \{x \in V : \|x\| \leq 1\}$



For any $x \in \mathbb{C}^n$, we have

$$\|x\|_\infty \leq \|x\|_2 \leq \|x\|_1$$

because

$$\|x\|_\infty = \sqrt{\max_k |x_k|^2} \leq \sqrt{\sum_k |x_k|^2}$$

$$x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}, \quad e_j = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \quad (1 \text{ in the } j\text{th slot})$$

where $(e_j)_i = S_{ij} = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$

So we can rewrite

$$\|x\|_2 = \|x_1 e_1 + x_2 e_2 + \cdots + x_n e_n\|_2$$

By the triangle inequality we have

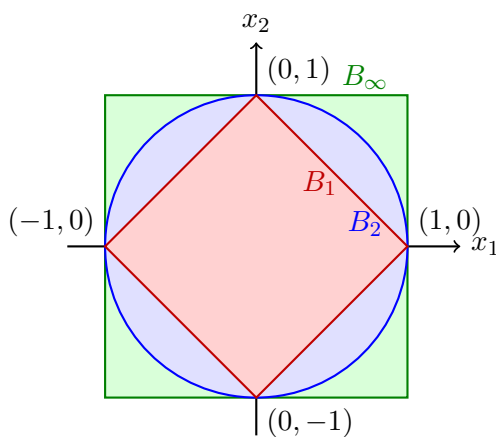
$$\|x_1 e_1 + x_2 e_2 + \cdots + x_n e_n\|_2 \leq \|x_1 e_1\|_2 + \|x_2 e_2\|_2 + \cdots + \|x_n e_n\|_2$$

$$\|x_1 e_1\|_2 + \cdots + \|x_n e_n\|_2 = |x_1| \|e_1\|_2 + |x_2| \|e_2\|_2 + \cdots + |x_n| \|e_n\|_2$$

Notice by definition of e_i its two norm would be 1, so this is just $\sum_i |x_k| = \|x\|_1$

Remark 1.3 (Unit Balls Nested in the opposite direction). We have

$$B_1 \subseteq B_2 \subseteq B_\infty$$



if $x \in B$, then $\|x\|_1 \leq 1$ so

$$\|x\|_2 \leq \|x\|_1 \leq 1$$

2 Inner Products

An inner product is a mapping $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{C}$ such that

1. $\langle u, u \rangle \geq 0$ if $u \neq 0$
2. $\langle v, u \rangle = \overline{\langle u, v \rangle}$
3. $\langle \alpha u, v \rangle = \alpha \langle u, v \rangle$.
4. $\langle u + v, w \rangle = \langle u, w \rangle + \langle v, w \rangle$
5. 3 implies that $\langle 0, v \rangle = 0$
6. 2 and 3 imply that $\langle u, \alpha v \rangle = \overline{\alpha} \langle u, v \rangle$
7. 2 and 4 imply that $\langle w, u + v \rangle = \langle w, u \rangle + \langle w, v \rangle$

Geometrically:

$$\langle u, v \rangle = \|u\| \cdot \|v\| \cdot \cos \theta$$

$$\|\alpha u\| = \sqrt{\langle \alpha u, \alpha u \rangle} = \sqrt{\alpha \overline{\alpha} \langle u, u \rangle} = |\alpha| \sqrt{\langle u, u \rangle}$$

In physics, usually the inner product is defined to be linear in y .

Theorem 2.1 (Cauchy–Schwarz) For all $u, v \in V$:

$$|\langle u, v \rangle| \leq \|u\| \cdot \|v\|$$

The proof – along with Pythagoras and the corollaries (triangle inequality, induced norm, angles) – is written out in the dedicated section at the end of these notes.

Definition 2.2 (Orthogonality). Vectors $e_1, e_2, \dots, e_n$ are orthogonal if $\langle e_j, e_k \rangle = S_{jk} = \begin{cases} 1, & j = k, \\ 0, & j \neq k. \end{cases}$

Recall that for an orthonormal basis, $e_1, e_2, \dots, e_n$ is that you can compute the coefficients with an inner product

$$x = \sum_{j=1}^n \alpha_j e_j = \langle \sum_j \alpha_j e_j, e_k \rangle = \sum_j \alpha_j \langle e_j, e_k \rangle$$

You can do Gram - Schmidt orthogonalization to create a new orthonormal basis given linearly independent vectors, and the orthogonal vectors will have the same span.

3 Definitions and notation

Definition 3.1 (First definition). Write the definition and one sentence explaining why it matters.

4 Proofs: Cauchy–Schwarz, Pythagoras, and Corollaries

This section is the dedicated write-up of the material flagged under the Cauchy–Schwarz theorem: the Cauchy–Schwarz proof, Pythagoras, and the corollaries (triangle inequality, induced norm,

angles). Textbook reference: Boggess & Narcowich, §0.4–0.5.

4.1 Cauchy–Schwarz inequality

Theorem 4.1 (Cauchy–Schwarz, full statement) *For all u, v in an inner product space V (real or complex),*

$$|\langle u, v \rangle| \leq \|u\| \cdot \|v\|.$$

Equality holds if and only if u and v are linearly dependent. Moreover, $\langle u, v \rangle = \|u\| \cdot \|v\|$ if and only if one of u, v is a nonnegative multiple of the other.

Idea. Make $\langle u, v \rangle$ real and nonnegative by rotating u by a phase, then expand $\|e^{-i\phi}u - tv\|^2 \geq 0$ as a quadratic in t and read the inequality off its discriminant.

Proof. If $v = 0$ or $\langle u, v \rangle = 0$ the inequality is immediate, so assume $v \neq 0$ and $\langle u, v \rangle \neq 0$. Write $\langle u, v \rangle = |\langle u, v \rangle|e^{i\phi}$. By linearity in the first argument,

$$\langle e^{-i\phi}u, v \rangle = e^{-i\phi}\langle u, v \rangle = |\langle u, v \rangle| \geq 0,$$

which is real. Hence for every real t ,

$$\begin{aligned} 0 \leq \|e^{-i\phi}u - tv\|^2 &= \langle e^{-i\phi}u - tv, e^{-i\phi}u - tv \rangle \\ &= \|u\|^2 - t\langle e^{-i\phi}u, v \rangle - t\langle v, e^{-i\phi}u \rangle + t^2\|v\|^2 \\ &= \|u\|^2 - 2t|\langle u, v \rangle| + t^2\|v\|^2, \end{aligned}$$

because $\langle v, e^{-i\phi}u \rangle = \overline{\langle e^{-i\phi}u, v \rangle} = |\langle u, v \rangle|$ is real too. The last line is a quadratic in t that is nonnegative for every real t ; since its leading coefficient $\|v\|^2$ is positive, its discriminant must be nonpositive:

$$4|\langle u, v \rangle|^2 - 4\|u\|^2\|v\|^2 \leq 0 \quad \implies \quad |\langle u, v \rangle| \leq \|u\| \cdot \|v\|.$$

Equality means the discriminant vanishes, so the quadratic has the real double root $t_0 = |\langle u, v \rangle|/\|v\|^2 \geq 0$; then $\|e^{-i\phi}u - t_0v\|^2 = 0$, i.e. $e^{-i\phi}u = t_0v$, so u and v are linearly dependent. Conversely, if $u = \alpha v$ then $|\langle u, v \rangle| = |\alpha|\|v\|^2 = \|u\| \cdot \|v\|$. Finally, $\langle u, v \rangle = \|u\| \cdot \|v\|$ (real and positive, so $\phi = 0$) holds exactly when $u = t_0v$ with $t_0 \geq 0$: one is a nonnegative multiple of the other. $\square$

4.2 Pythagorean theorem

Theorem 4.2 (Pythagoras) *If $\langle u, v \rangle = 0$, then*

$$\|u + v\|^2 = \|u\|^2 + \|v\|^2.$$

More generally, if $e_1, \dots, e_N$ are orthonormal and $v = \sum_{j=1}^N \alpha_j e_j$, then

$$||v||^2 = \sum_{j=1}^N |\alpha_j|^2.$$

Proof. Expand and use conjugate symmetry:

$$||u+v||^2 = \langle u+v, u+v \rangle = ||u||^2 + \langle u, v \rangle + \langle v, u \rangle + ||v||^2 = ||u||^2 + 2 \operatorname{Re} \langle u, v \rangle + ||v||^2.$$

If $\langle u, v \rangle = 0$ both cross terms vanish, which is the first claim. For the general version,

$$||\sum_{j=1}^N \alpha_j e_j||^2 = \sum_{j,k=1}^N \alpha_j \overline{\alpha_k} \langle e_j, e_k \rangle = \sum_{j,k=1}^N \alpha_j \overline{\alpha_k} \delta_{jk} = \sum_{j=1}^N |\alpha_j|^2.$$

This identity is what becomes Parseval's equation later in the course. □

4.3 Corollaries

Corollary 4.3 (Triangle inequality) For all $u, v \in V$,

$$||u+v|| \leq ||u|| + ||v||.$$

Proof. Using the expansion from the Pythagoras proof and then Cauchy-Schwarz,

$$||u+v||^2 = ||u||^2 + 2 \operatorname{Re} \langle u, v \rangle + ||v||^2 \leq ||u||^2 + 2|\langle u, v \rangle| + ||v||^2 \leq ||u||^2 + 2||u|| \cdot ||v|| + ||v||^2 = (||u|| + ||v||)^2.$$

Both sides are nonnegative, so taking square roots gives the inequality. □

Remark 4.4 (Every inner product induces a norm). This closes the loop with the norm definition at the top of the lecture: $||u|| = \sqrt{\langle u, u \rangle}$ is a genuine norm. Positive definiteness is axiom 1 of the inner product, homogeneity was checked in the Inner Products section ($||\alpha u|| = |\alpha| ||u||$), and the triangle inequality is exactly theorem 4.3. On $\mathbb{C}^n$ with the standard inner product, this induced norm is the two-norm.

Remark 4.5 (Angles are well defined). For nonzero vectors, Cauchy-Schwarz gives $|\langle u, v \rangle| / (||u|| \cdot ||v||) \leq 1$, so in a real inner product space we can define the angle θ between u and v by

$$\cos \theta = \frac{\langle u, v \rangle}{||u|| \cdot ||v||}.$$

This is what justifies the geometric formula $\langle u, v \rangle = ||u|| \cdot ||v|| \cdot \cos \theta$ from earlier in the lecture; orthogonality ($\langle u, v \rangle = 0$) is exactly $\theta = \pi/2$. In a complex space, use $\operatorname{Re} \langle u, v \rangle$ in the cosine.

Example 4.6 (Cauchy–Schwarz in the standard spaces). • On $\mathbb{C}^n$ with the standard inner product:

$$\left| \sum_{j=1}^n x_j \overline{y_j} \right| \leq \left(\sum_{j=1}^n |x_j|^2 \right)^{1/2} \left(\sum_{j=1}^n |y_j|^2 \right)^{1/2}.$$

- On $L^2[a, b]$, where $\langle f, g \rangle = \int_a^b f(t) \overline{g(t)} dt$:

$$\left| \int_a^b f(t) \overline{g(t)} dt \right| \leq \left(\int_a^b |f(t)|^2 dt \right)^{1/2} \left(\int_a^b |g(t)|^2 dt \right)^{1/2}.$$

This integral version is the workhorse inequality for the rest of the course (Fourier series, convergence in L^2 , Parseval).

5 Examples and calculations

Example 5.1 (Lecture example). Write the setup, the calculation, and the conclusion.

6 Questions and follow-up

- What is still unclear?
- Which exercise or reading should I revisit?
- TODO: